MILITARY SPECIFICATION

MOLDING MATERIAL, PLASTIC, EPOXY COMPOUNDS

THERMOSETTING

1. SCOPE

1.1 Scope. This specification covers the basic properties of thermosetting molding material, plastic, epoxy compounds, as such, and the methods suited to their satisfactory determination.

1.2 Classification. Molding thermosetting epoxy compounds shall be of the following types, as specified (see 6.2).

Type MEC - Mineral filler, encapsulation grade characterized by its ability to flow and mold at extremely low pressure and to flow over or around inserts without displacing or distorting them in a mold.
Type MEE - Mineral filler, low pressure, electrical grade; also can be used for encapsulation purposes.
Type MEG - Mineral filler, general purpose.
Type MEH - Mineral filler, heat resistant.
Type GEI-5- Glass fiber filler, impact strength 0.5 foot-pounds per inch notch.
Type GEI-20- Glass fiber filler, impact strength 2.0 foot-pounds per inch notch.
Type GEI-100-Glass fiber filler, impact resistant, impact strength 10.0 foot-pounds per inch notch.

2. APPLICABLE DOCUMENTS

2.1 This section is not applicable to this specification

2.2 Other publications. - The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 149 - Dielectric Breakdown Voltage and Dielectric Strength of Electrical Insulating Materials at Commercial Power Frequencies, Tests for.
D 150 - A-C Loss Characteristics and Dielectric Constant (Permitivity of Solid Electrical Insulating Materials, Tests for.
D 229 - Rigid Sheet and Plate Materials Used for Electrical Insulation, Testing.
D 256 - Impact Resistance of Plastics and Electrical Insulating Materials, Tests for.
D 495 - High-Voltage, Low Current Arc Resistance of Solid Electrical Insulating Materials, Test for.
D 570 - Water Absorption of Plastics, Test for.
D 618 - Conditioning Plastics and Electrical Insulating Materials for Testing.
D 648 - Deflection Temperature of Plastics Under Load, Test for.
D 651 - Tensile Strength of Molded Electrical Insulating Materials, Test for.
D 695 - Compressive Properties of Rigid Plastics, Test for.
D 790 - Flexural Properties of Plastics, Test for.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Qualification. Thermosetting molding materials furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.3 and 6.3)

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3.2 Compound. The compound shall consist of a suitable filler impregnated or intimately combined with a thermosetting polymerization product binder, processed to conform to this specification. No scrap compound previously cured and reground shall be utilized.

3.2.1 Uniformity. All molding compound of the same brand from one manufacturer shall be uniform in texture, in color (see 3.2.2), and in the specified properties as determined by the batch-acceptance inspection specified in 4.4.

3.2.2 Color. The compounds shall be supplied in the color specified by the procuring activity (see 6.2 and 6.5). If the color is not specified, the compound shall be furnished in the normal color in which manufactured. Natural-color compounds are those in which no coloring matter of any sort has been added. In evaluating uniformity of color, consideration shall be given to the fact that variation in color of raw materials, particularly in the filler, may be reflected in the color of the natural molding compound. Coloring matters reducing the electrical properties below the specified limits shall not be used.

3.2.3 Property values. Standard specimens of the compounds shall conform to the property values specified in tables I, II, III.

3.3 Workmanship. The compound manufactured and processed in accordance with this specification shall be such as to meet all the requirements of this specification and any referenced subsidiary specification, drawing or other document when inspected in accordance with section 4.

Table I - Property values of plastic molding epoxy compound.

PROPERTY TO BE TESTED		VALUES REQUIRED FOR EACH TYPE OF COMPOUND						
		TYPE MEC	TYPE MEE	TYPE MEG	TYPE MEH	TYPE GEI-5	TYPE GEI-20	TYPE GEI-100
ARC RESISTANCE		120	180	150	150	150	130	130
COMPRESSIVE STRENGTH (ENDWISE)		15000	20000	20000	20000	25000	20000	20000
DIELECTRIC CONSTANT								
1 KILOCYCLE	DRY	5.2	5.0	5.3	5.8	5.8	5.8	5.8
	WET	5.4	5.2	5.5	6.0	6.0	6.0	6.0
1 MEGACYCLE	DRY	5.0	4.8	5.5	5.5	5.5	5.5	5.5
	WET	5.2	5.0	5.7	5.7	5.7	5.7	5.7
DISSIPATION FACTOR								
1 KILOCYCLE	DRY	0.015	0.015	0.020	0.025	0.025	0.025	0.025
	WET	0.020	0.020	0.025	0.030	0.030	0.030	0.030
1 MEGACYCLE	DRY	0.012	0.012	0.025	0.020	0.020	0.020	0.020
	WET	0.015	0.015	0.030	0.025	0.025	0.025	0.025
DIELECTRIC STRENGTH								
SHORT TIME	DRY	325	325	325	325	325	300	300
STEP BY STEP		325	325	325	325	325	300	300
SHORT TIME	WET	300	300	300	300	300	275	275
STEP BY STEP		300	300	300	300	300	275	275
DIELECTRIC BREAKDOWN								
SHORT TIME	DRY	40	40	40	40	40	40	40
STEP BY STEP		35	35	35	35	35	35	35
SHORT TIME	WET	40	40	40	40	40	40	40
STEP BY STEP		35	35	35	35	35	35	35
FLAME RESISTANCE								
IGNITION TIME		90	90	90	90	90	90	90
BURNING TIME		70	70	70	70	70	70	70

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Table I - Property values of plastic molding epoxy compound—Continued.

PROPERTY TO BE TESTED	VALUES REQUIRED FOR EACH TYPE OF COMPOUND						
	TYPE MEC	TYPE MEE	TYPE MEG	TYPE MEH	TYPE GEI-5	TYPE GEI-20	TYPE GEI-100
FLEXURAL STRENGTH (FACE)	8000	10000	10000	12000	15000	16000	16000
HEAT-DISTORTION TEMP. (SIDE)	100	130	130	175	130	150	175
IMPACT STRENGTH							
FACE	0.2	0.3	0.3	0.35	0.50	2.0	10.0
SIDE	0.2	0.3	0.3	0.35	0.50	2.0	10.0
TENSILE STRENGTH	3500	7000	6000	7000	8000	8000	8000
WATER ABSORPTION	0.5	0.20	0.30	0.30	0.30	0.50	0.50
HEAT RESISTANCE	--	--	--	25	--	--	--
DIMENSIONAL STABILITY (HIGH TEMP.)	--	0.1	0.1	0.1	0.1	0.1	0.2
VOLUME RESISTANCE	1.0	1000	5.0	100	1000	1000	1000
SURFACE RESISTANCE	1.0	1000	5.0	100	1000	1000	1000

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. The tests shall be classified as follows:

(a) Qualification tests (see 4.3).
(b) Batch-acceptance inspection (see 4.4).

4.3 Qualification tests. 1/ Qualification tests shall be conducted at a laboratory satisfactory to the Naval Ship Engineering Center. Qualification tests shall consist of all the tests specified in table II (see 4.5).

4.3.1 Samples for qualification tests. Tests shall be conducted on standard test specimens molded of each type compound as specified in table II. Approval of one color will include approval for all colors (see 3.2.2).

4.3.1.1 Quantities to be furnished. The quantity of standard test specimens to be furnished shall be as specified in table II for each compound.

4.3.1.2 Preparation. Specimens shall be prepared for tests as specified in 4.5.1.

4.4 Batch acceptance inspection. Batch acceptance inspection shall be as specified in table III. Tests shall be conducted on standard specimens molded by the manufacturer from each batch of compound. The molding of specimens and conduct of tests shall be under the surveillance of the Defense Contract Administration Service (DCAS) representative.

4.4.1 Definition. A batch is a homogeneous unit of finished molding compound manufactured at one time.

4.4.2 Preparation of specimens. Specimens shall be prepared as specified in 4.5.1.

4.4.3 Results of tests. One copy of the results of the batch acceptance inspection shall be furnished to the Defense Contract Administration Service (DCAS), and one copy of the test report shall be forwarded to Code 936, U.S. Naval Applied Science Laboratory, Flushing and Washington Avenues, Brooklyn, New York 11251

1/ Application for Qualification tests shall be made in accordance with "Provisions Governing Qualification SD-6" (See 6.3 and 6.4).

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Table II - Test procedures for qualification tests

Property to be tested	Test methods: Specification	Test methods: Modified by	Specimens: Form and dimensions	Specimens: Number tested	Conditioning Procedure	Units
Arc resistance	D 495	Tungsten electrodes	4 inch disk, 0.125±0.010 inch in thickness	3[3/]		Seconds (minimum avg.)
Compressive strength endwise	D 695		1 by 1/2 by 1/2 inch	5	48/50+96/23/50	P. S. I (minimum average)
Dielectric constant				3	48/50+des	
At 1 kilocycle	D 150		4 inch disk, 1/8 inch thick	3	48/50+24/23 distilled water	(Maximum average)
At 1 megacycle	D 150		2 inch disk, 1/8 inch thick	3	48/50+des	
				3	48/50+24/23 distilled water	
Dissipation factor				3	48/50+des	
At 1 kilocycle	D 150		4 inch disk, 1/8 inch thick	3	48/50+24/23 distilled water	(Maximum average)
				3	48/50+des	
At 1 megacycle	D 150		2 inch disk, 1/8 inch thick	3	48/50+24/23 distilled water	
Dialectric strength						
Short time test			4 inch disk, 1/8 inch thick	3	48/50+96/23/50	
Step by step test	D 149	Under Transformer oil		5		Volts per mil (minimum average)
Short time test	D 149	"	4 inch disk, 1/8 inch thick	3	48/50+48/50 distilled water	
Step by step test				5		
Dielectric breakdown						
Short time test				1	48/50+96/23/50	
Step by step	D 149	4. 5. 2. 1. 3	See figure 1 herein	3		Kilovolts (minimum average)
Short time test	D 149	4. 5. 2. 1. 3		1	48/50+48/50 distilled water	
Step by step test				3		
Flame resistance						
Ignition time	D 229	Method 2	5 inch bar, 1/2 by 1/2 inch	5		Seconds (minimum average)
Burning time						Seconds (maximum average)
Flexural strength						
Face[1/]	D 790	Procedure A	5 inch bar, 1/4 by 1/2 inch	5	48/50+96/23/50	P. S. I. (minimum average)
Heat distortion						
Temperatures, side[2/]	D 648		5 inch bar, 1/2 by 1/2 inch	3		°C (minimum average)

See footnotes at end of table II.

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Table II - Test procedures for qualification tests–Continued

Property to be tested	Test methods Specification	Modified by	Specimens Form and dimensions	Number tested	Conditioning Procedure	Units
Impact strength Face[1/] Side[2/]	D 256	Method A	See figure 2 of ASTM D256	5 5	48/50+96/23/50	Foot-pounds per inch notch (minimum average)
Tensile strength	D 651		See figure 2 of ASTM D651	5	48/50+96/23/50	P.S.I. (minimum average)
Water absorption	D 570		2 inch disk, 1/8 inch thick	5	24/100+des+48/50 distilled water	Percent (maximum average)
Heat resistance	D 790	4.5.2.5	5 inch bar, 1/4 by 1/2 inch	5	48/50+96/23/50+1/200	Percent flexural strength retained (minimum)
Dimensional stability High temperature		4.5.2.6	5 inch bar, 1/2 by 1/2 inch	5	See 4.5.2.6	Percent (maximum average)
Volume resistance		4.5.2.7	4 inch disk, 1/8 inch thick	5	see 4.5.2.7	Megohms (minimum) (individual)
Surface resistance		4.5.2.7	4 inch disk, 1/8 inch thick	5	see 4.5.2.7	Megohms (minimum) (individual)

1/ The face of a test specimen is that area formed by the top or bottom force plug
2/ The side of a test specimen is that area formed by the chase of the mold
3/ Five readings shall be made on each of the three specimens

Table III - Test procedures and property values for batch acceptance tests

Property to be tested	Test method Specification	Modified by	Specimens Form and dimensions	Number tested	Conditioning procedures	Units
Arc resistance	D 495	Tungsten electrodes	4 in. disk, 0.125±0.010 inch in thickness	3		Seconds (minimum average)
Dielectric constant at 1 megacycle	D 150		2 in disk, 1/8 inch thick	3	48/50+24/23 distilled water	(Maximum average)
Dissipation factor at 1 megacycle	D 150		2 inch disk, 1/8 inch thick	3	48/50+24/23 distilled water	(Maximum average)

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Table III - Test procedures and property values for batch acceptance tests—Continued

Property to be tested	Test method Specification	Modified by	Specimens Form and dimensions	Number tested	Conditioning procedure	Units
Dielectric strength Step-by-step	D 149		4 inch disk, 1/8 inch thick	5	48/50+48/50 distilled water	Volts per mil (minimum average)
Flexural strength Face[1/]	D 790		5 inch bar, 1/4x1/2 inch	5	48/50+96/23/50	P. S. I. (minimum average)
Impact strength Side[2/]	D 256	Method A	See fig 2 of ASTM D256	5	48/50+96/23/50	Foot-pounds per inch notch (minimum average)
Water absorption	D 570		2 inch disk, 1/8 inch thick	5	24/100+des+48/50 distilled water	Percent (maximum average)

Property to be tested	Value required for each type of compound						
	Type MEC	Type MEE	Type MEG	Type MEH	Type GEI-5	Type GEI-20	Type GEI-100
Arc resistance	120	180	150	150	150	130	130
Dielectric constant at 1 megacycle, wet	5.2	5.0	5.7	5.7	5.7	5.7	5.7
Dissipation factor at 1 megacycle, wet	0.015	0.015	0.030	0.025	0.025	0.025	0.025
Dielectric strength Step-by-step	300	300	300	300	300	275	275
Flexural strength Face[1/]	8,000	10,000	10,000	12,000	15,000	16,000	16,000
Impact strength Side[2/]	0.2	0.3	0.3	0.35	0.50	2.0	10.0
Water absorption	0.5	0.20	0.30	0.30	0.30	0.50	0.50

1/ The faces of a test specimen is that area formed by the top or bottom force plug.
2/ The side of a test specimen is that area formed by the chase of the mold.

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4.4.4 Rejection. Failure to comply with table III shall be cause for rejection of the batch of compound.

4.5 Test procedures.

4.5.1 Standard test specimens.

4.5.1.1 Number. The minimum number of standard test specimens to be tested shall be as specified in tables II and III.

4.5.1.2 Form. The form of the standard test specimens shall be as specified in the applicable methods and as specified in tables II and III.

4.5.1.3 Molding of test specimens. Test specimens shall be molded by methods which are representative of sound practice. No special treatment shall be used to improve the properties of the specimens.

4.5.1.4 Tolerance. Tolerance on dimensions shall be plus or minus 5.0 percent or as otherwise specified in test method.

4.5.1.5 Conditioning. Standard test specimens shall be conditioned before test, as specified in tables II and III.

4.5.1.5.1 Nomenclature. The designations indicating conditioning of test specimens shall be in accordance with ASTM D 618.

4.5.2 Methods of tests. All test measurements shall be taken at standard laboratory atmosphere of 23°C+2°C and 50 percent plus or minus 5 percent relative humidity with the exception of dielectric strength, dielectric breakdown, heat resistance, and volume and surface resistance (see 4.5.2.1, 4.5.2.1.3, 4.5.2.5, and 4.5.2.7). The test methods shall be in conformance with the applicable method of this specification, tables II and III, modified as follows:

4.5.2.1 Dielectric Strength. Step-up transformer, circuit breaker, voltage control and voltmeter shall be in accordance with ASTM D149. Power supply frequency shall not be greater than 100 hertz. The transformer shall have a rating of not less than 2 KVA. Tests shall be made under oil with electrodes 1 inch in diameter and 1 inch in length with edges rounded to 1/8 inch radius.

4.5.2.1.1 Short-time test. The voltage shall be increased uniformly at the rate of 500 volts per second.

4.5.2.1.2 Step-by-step test. The voltage shall be increased in increments, as specified in table IV up to failure, and shall be held at each step for one minute. The change from one step to the next higher shall be made within ten seconds.

Table IV - Voltage increase for step-by-step test

Breakdown by short-time method	Increments of increase
Kilovolts	Kilovolts
12.5 or less	0.5
Over 12.5 to 25, inclusive	1.0
Over 25 to 50, inclusive	2.5
Over 50 to 100, inclusive	5.0

4.5.2.1.3 Dielectric breakdown. Step-up transformer, circuit breaker, voltage control and voltmeter shall be in accordance with ASTM D 149. Power supply frequency shall not be greater than 100 hertz. The transformer shall have a rating of not less than 2 KVA. Tests shall be made under oil with the specimen having electrodes of American Standard No. 3 tapered pins. The test potential shall be applied successively between the numbered pairs of electrodes (see figure 1) and the average of three readings shall be taken as the reading for the specimen.

4.5.2.2 Flexural strength. Flexural strength test shall be in accordance with procedure A of ASTM D 790. The span-depth ratio shall be 16 to 1.

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4.5.2.3 Heat distortion temperature. Heat distortion temperature shall be determined in accordance with ASTM D 648 with the specimens loaded to a fiber stress of 264 psi ± 2-1/2 percent. The specimens shall be placed directly in the oil bath and not in an air bath surrounded by an oil bath.

4.5.2.4 Water absorption. Water absorption shall be determined in accordance with ASTM D 570, with the following modifications:

(a) The specimens shall be conditioned at 100°C ± 2°C. for twenty four hours, followed by 16 to 20 hour period of cooling over silica gel or calcium chloride in a dessicator at 23°±1.1°C.

(b) The specimens shall be immersed in distilled water maintained at a temperature of 50°C±1°C for forty eight hours. The report shall include only the following data: The percentage increase in weight during immersion calculated to the nearest 0.01 percent as follows:

$$\text{Increase in weight (percent)} = \frac{\text{wet weight - conditioned weight}}{\text{conditioned weight}} \text{ x100}$$

4.5.2.5 Heat resistance. The method shall be as specified in 4.5.2.2 with conditioning as specified in table II. The test shall be conducted at 200°C.

The average of five such determinations divided by the average flexural strength determined as in 4.5.2.2 shall be multiplied by 100 and recorded as percent flexural strength retained.

4.5.2.6 Dimensional stability (high temperature). The specimens shall be machined so the 1/2 by 1/2 inch ends are smooth and parallel. The specimens shall be subjected to condition 96/23/50, (see 4.5.1.5.1). The initial length of the specimens shall then be measured to the nearest 0.001 inch. The specimens shall then be subjected to 10 cycles, each cycle as follows: forty eight hours in a circulating air oven at 125°±5°C. plus twenty four hours at 23°C±1°C and 50 percent ± 2 percent relative humidity. At the completion of ten cycles the final length of the specimens shall be measured to the nearest 0.001 inch. The percentage dimensional change is calculated to the nearest 0.1 percent as follows:

$$\text{Dimensional change (percent)} = \frac{\text{(initial length - final length)}}{\text{initial length}} \text{ x 100}$$

The average-percent dimensional change of the five specimens shall be recorded.

4.5.2.7 Volume and surface resistance. The specimens shall be subjected to condition 720/70/100+dew.

4.5.2.7.1 Specimens. Four inch diameter, 1/8 inch thick specimens shall be used. Specimens shall be cleaned by noninjurious methods to assure freedom from contamination. Precautions shall be taken in handling the specimens to avoid additional contamination.

4.5.2.7.2 Electrodes. Electrodes shall consist of a guarded electrode 2 inches in diameter, 1/4 inch guard ring spaced 1/4 inch from the guarded electrode on the same side and the third electrode 3 inches in diameter on the opposite side and concentric with the guarded electrode. Dimensions of electrodes shall be maintained to a tolerance of plus or minus 1/64 inch. Silver paint, permeable to moisture (equivalent to Dupont No. 4817) shall be used for painting electrodes on the specimens. The electrodes shall exhibit a resistance of not more than 6 ohms both before and after the 720/70/100 + dew conditioning when measured with potentials of not greater than 3 volts between points diametrically opposite on each electrode. The specimens shall be permitted to air dry after painting for at least one week in an atmosphere of less than 60 percent relative humidity at a temperature of 23°C±2°C.

4.5.2.7.3 Humidity chamber. The humidity chamber shall consist of a glass container with a corrosion resistant cover. The cover shall be provided with thru-panel type insulators. The insulators may serve as supports for the electrode holders as shown on figure 2. The chamber shall be of such size that the ratio of specimens surface area to water surface area shall not exceed 2.5. The ratio of volume of air in the humidity chamber to surface area of the water shall not exceed 10. One hundred percent relative humidity with condensation shall be obtained by natural evaporation from a quantity of distilled water located at the bottom of the chamber. The cover shall be sealed to the chamber by an inert sealing compound applied to the exterior joints formed by the cover and the walls of the chamber. A small vent hole shall be provided in the cover to equalize pressure. The vent hole shall be sealed as soon as the air temperature in the humidity chamber has reached 70°C.

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4.5.2.7.4 Specimen holders. The specimens shall be installed in a vertical plane in the conditioning chamber with the lower edge of the specimen not closer than 1 inch from the surface of the water. The specimens may be held in position by the electrode contactors in a manner similar to that shown on figure 2. Electrical connection to the specimen holders may be made by means of the thru - panel insulators. The insulators shall be capable of with-standing the adverse conditions within the chamber without excessive loss of insulating properties. (Insulator resistance to cover plate shall at all times exceed 10 megohms). Polytetrafluoroethylene insulators on the humidity side of the conditioning chamber are recommended to meet this requirement. These should be cleaned with alcohol before the start of each test. Electrode contactors and all other metallic parts of the sample shall be silver plated. Contact pressure against the electrodes may be provided by backing the contactors with phosphor bronze springs or other corrosion-resistant spring material.

4.5.2.7.5 Heating chamber. The humidity chamber shall be installed in an oven or other heating chamber capable of maintaining a temperature of 70°C±1°C. The rate of heating of the oven shall be such that the air temperature at a point near the volumetric center of the humidity chamber shall attain 70°C in 4 hours plus or minus 1 hour. The quantity of water in the chamber shall be such that the water temperature shall attain 65°C in 4 hours plus or minus 1 hour. Room temperature shall be maintained at 25°C±5°C. The insulator of the conductors connecting the through-panel insulation to the measuring equipment shall not be significantly deteriorated by the elevated temperatures encountered in the oven. Polytetrafluoroethylene coated wire is recommended.

4.5.2.7.6 Measurements. 1/ Volume and surface resistances shall be measured by the three terminal method employing measuring equipment such as a megohm bridge capable of applying 500 volts direct current (d.c.) to the specimen. A single set of measurements shall be made of each specimen while in the conditioning chamber after 30 days of the specified conditioning. Conversion of measurements to resistivities is not required since electrode dimensions are specified. The potentials shall be applied to the specimens as shown on figure 3 or with polarities opposite to those shown on the figure. Surface resistance measurements shall be made on the same specimens as those used for volume resistance except that the potentials of guard and low electrodes shall be interchanged. The volume and surface resistance shall be measured in each case, 1 minute after the potentials are applied. Low values of volume and surface resistance (below 5 megohms) may be measured by the circuits shown on figure 4.

5. PREPARATION FOR DELIVERY

5.1 This section is not applicable to this specification.

6. NOTES

6.1 Intended Use.

6.1.1 Type MEC. This type is a mineral-filled compound intended for encapsulation purposes with good flow characteristics.

6.1.2 Type MEE. This type is a mineral-filled compound for use where good dielectric properties and arc and flame resistance are required.

6.1.3 Type MEG. This type is a mineral-filled, general-purpose, compound with good electrical and mechanical properties.

6.1.4 Type MEH. This type is a mineral-filled compound intended for applications requiring highest heat resistance.

1/ Because of the variability of the resistance of a given specimen with test conditions and because of non-uniformity of the same material from specimen to specimen, determinations are usually not reproducible to closer than 10 percent and are often even more widely divergent. A range of values of 10 to 1 may be obtained under apparently identical conditions. Errors in resistance determinations may result from the fact that the current measuring device is shunted by the resistance between the guarded terminal and the guard system. In some bridge techniques a standard resistor in the bridge is shunted by the resistance between the unguarded terminal and the guard system. The assure validity of the volume and surface resistance measurements obtained by the bridge methods the resistance between the unguarded terminal and the guard terminal should be at least five times greater than the standard resistance employed in the bridge. This may be ascertained by direct two terminal measurements between these two terminals.

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6.1.5 Type GEI-5. This type is a glass-filled compound of moderate impact strength.

6.1.6 Type GEI-20. This type is a glass-filled compound of high impact strength.

6.1.7 Type GEI-100. This type is a glass-filled compound of high impact strength.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type required (see 1.2)
(c) Color required (see 3.2.2).

6.3 With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in applicable Qualified Products List QPL 24325 whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Naval Ship Engineering Center, Department of the Navy, Washington, D.C. 20360, and information pertaining to qualification of products may be obtained from that activity. Application for Qualification tests shall be made in accordance with "Provisions Governing Qualification SD-6" (see 6.4).

6.4 Copies of "Provisions Governing Qualification SD-6" may be obtained upon application to Commanding Officer, Naval Supply Depot, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.

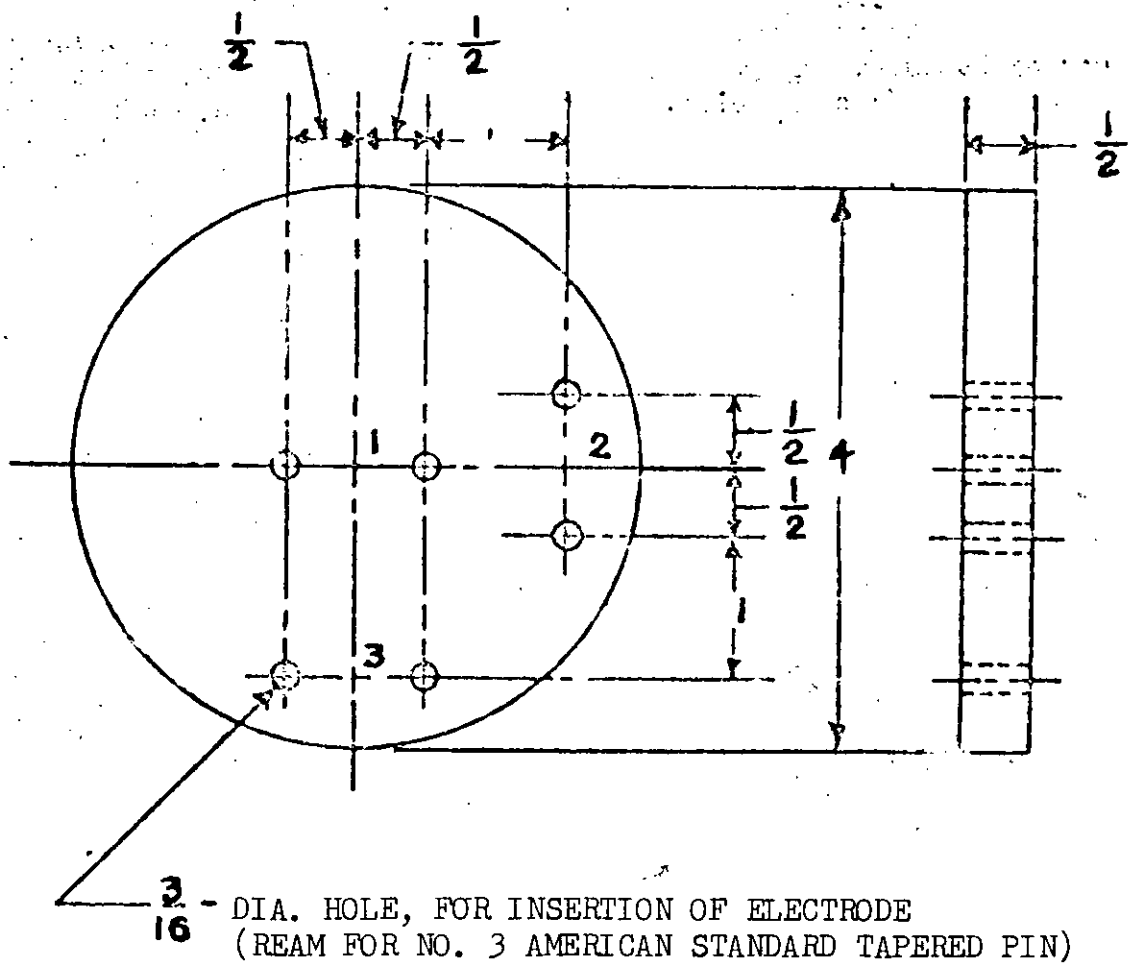
6.5 Toxicity when heated. It is desired that the following maximum concentrations of the toxic gases not be exceeded when the materials are tested in accordance with the methods of Defense Documentation Center test report No. 297457 (U.S. Testing Report No. 83413). Copies are available from the Defense Documentation Center (For Scientific and Technical Information), Cameron Station, Alexandria, Virginia 22314.

Toxic gas	Maximum parts per million
Carbon dioxide	15,000
Carbon monoxide	1,000
Ammonia	2,500
Aldehydes as H. CHO	50
Cyanide as HCN	60
Oxides of Nitrogen as NO_2	100
Hydrogen chloride	100

6.6 Color card. Color cards may be obtained upon application to the Chemical Laboratory, Norfolk Naval Shipyard, Portsmouth, Virginia 23709. The purpose for which cards are desired should be specified.

Preparing activity:
Navy - SH
(Project 9330-N397)

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NOTES:

1. ALL DIMENSIONS IN INCHES.
2. TOLERANCES ON DIMENSIONS, PLUS OR MINUS 5 PERCENT.
3. DISKS SHALL BE FURNISHED UNDRILLED AND SHALL BE DRILLED BY LABORATORY.

TEST: DIELECTRIC BREAKDOWN

FIGURE 1. Standard test specimen drilled for three pairs of electrodes.

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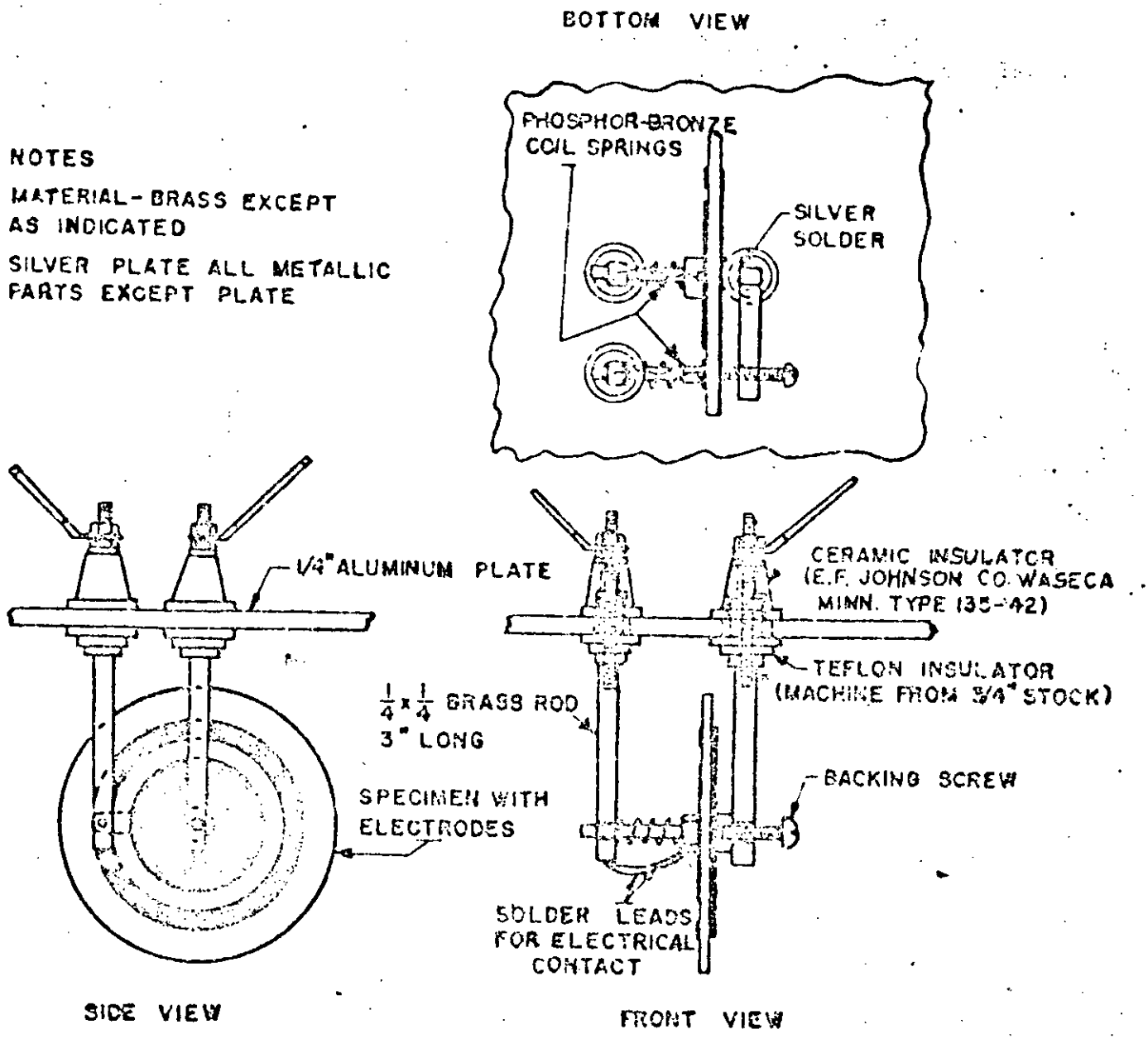


FIGURE 2. Specimen holders, electrodes, test samples, and humidity chamber cover — volume and surface resistance test.

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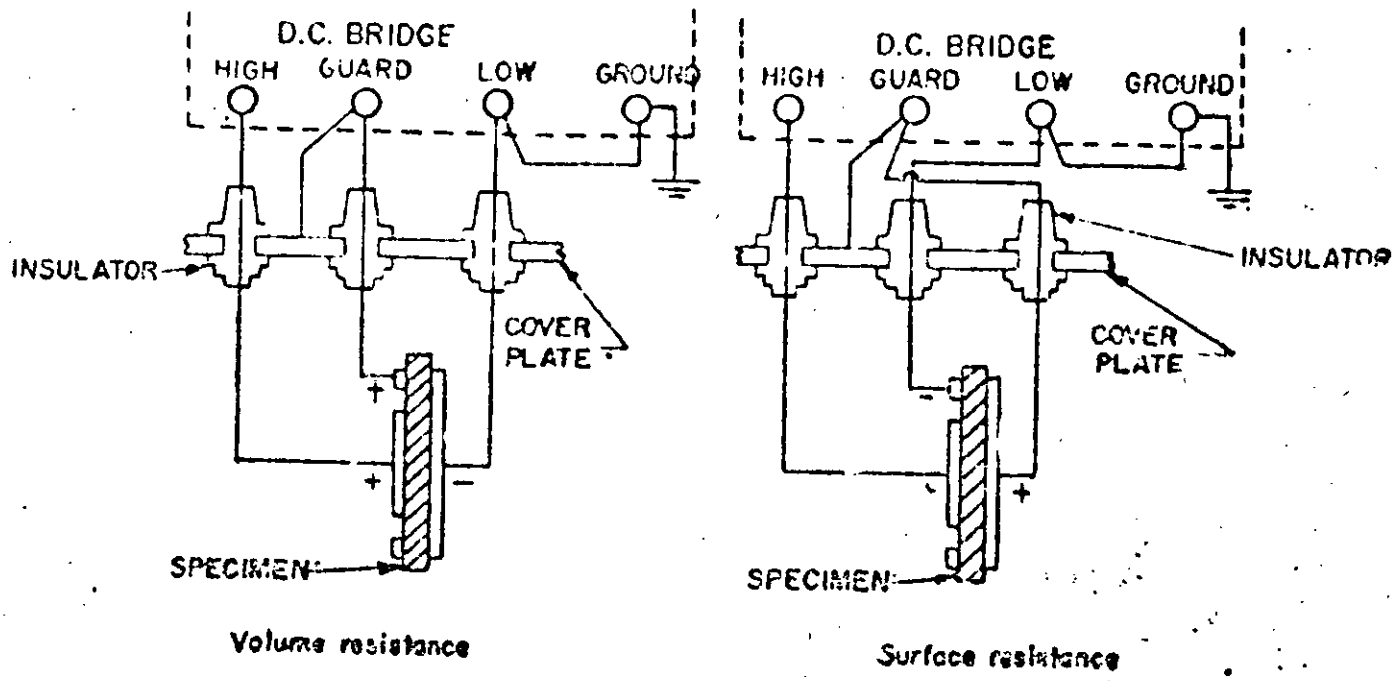
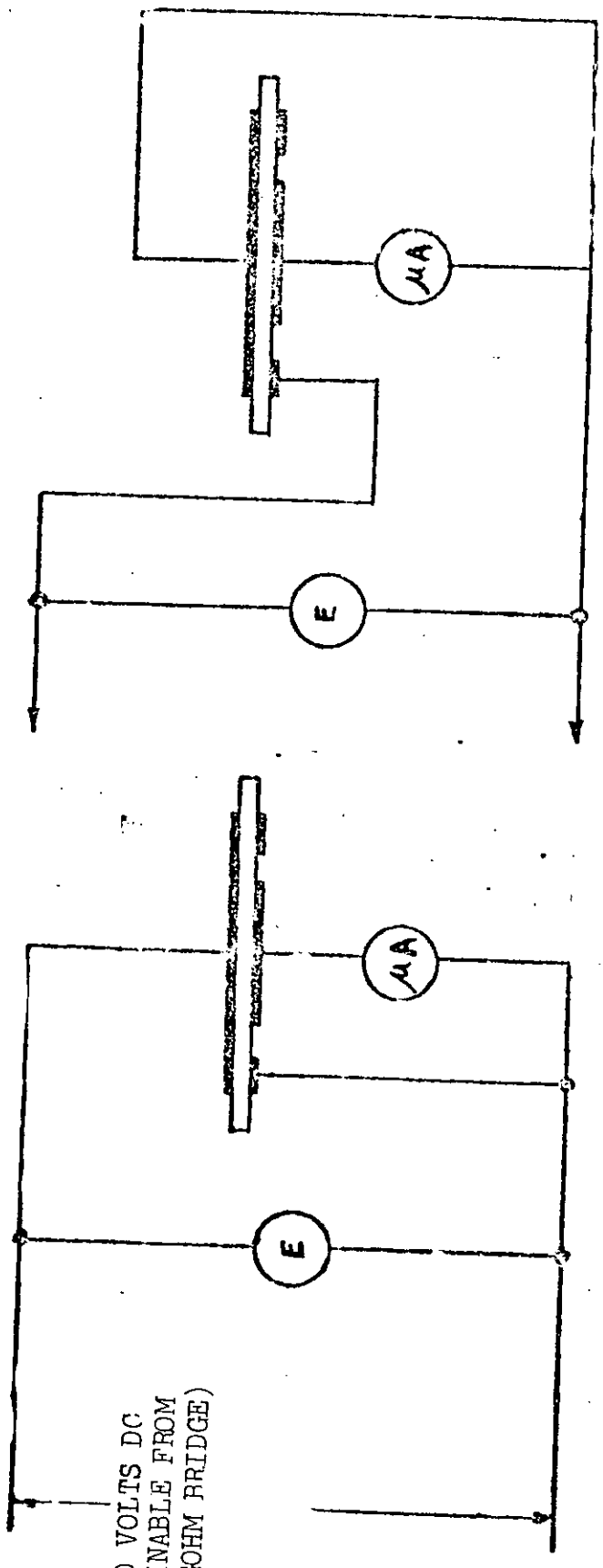


FIGURE 3. Arrangements for volume resistance and surface resistance test.

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$$R(\text{MEGOHMS}) = \frac{E}{\mu A}$$

FIGURE 4. Circuits for measuring low values of volume and surface resistance.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse of)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

(NAVSHIPS OVPRT 12-66)

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